

EXERCISE 1: SQL FUNDAMENTALS

(1) SELECT *
FROM employees;

OUTPUT TABLE

Id	first name	last name	department	Salary	hire date
1	Alice	Green	IT	70000	2020-01-10
2	Brain	Lee	HR	45000	2019-03-22
3	Cathy	Zulu	Finance	65000	2018-07-18
4	David	Mokocna	Marketing	50000	2021-11-05
5	Eva	Naidoo	IT	72000	2017-09-30

City
Johannesburg
Cape Town
Durban
Pretoria
Johannesburg

2) SELECT DISTINCT department
FROM employees;

OUTPUT TABLE

department
IT
HR
Finance
Marketing

3)

```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;
```

OUTPUT TABLE

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brain	Lee

(4)

```
SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

OUTPUT TABLE

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

(2) Filtering with WHERE

5) SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';

OUTPUT TABLE

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6) SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance'
AND salary > 60000;

OUTPUT

id	first_name	last_name	department
3	Cathy	Zulu	Finance

(7) SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';

OUTPUT

id	first_name	last_name	department
2	Brian	Lee	HR HR
4	David	Mokocni	Marketing

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main results of the paper. The second part is devoted to a discussion of the

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8) SELECT id, first_name, last_name, department
FROM employee

~~WHERE department = 'IT';~~
WHERE department NOT IN ('IT');

OUTPUT

id	first_name	last_name	department
2	Brain	Lee	HR
3	Cathy	Zulu	Finance
4	David	Makoeana	Marketing

(9) SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

OUTPUT

id	first_name	last_name	department
1	Alice	Green	IT
2	Brain	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10) SELECT id, first_name, last_name, department, salary, city
FROM employees

WHERE department = 'IT'

AND salary > 65000

AND city = 'Johannesburg';

OUTPUT

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

